

**MACHINE LEARNING BASED LAND USE AND LAND COVER CLASSIFICATION AND ZONING
USING SENTINEL IMAGERY OF HYDERABAD REGION**

A PROJECT REPORT

SUBMITTED TO



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LIST OF ABBREVIATIONS

AOI	Area of Interest
CRS	Coordinate Reference System
GEE	Google Earth Engine
GIS	Geographic Information System
KAN	Kolmogorov-Arnold Network
LULC	Land Use and Land Cover
MLE	Maximum Likelihood Estimation
MLP	Multi-Layer Perceptron
NDBI	Normalized Difference Built-up Index
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
QGIS	Quantum Geographic Information System
RF	Random Forest
RS	Remote Sensing
SNAP	Sentinel Application Platform
SVM	Support Vector Machine

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1. INTRODUCTION

LULC information with reliable, up-to-date and accurate analysis is crucial for informed decision-making in various sectors such as agriculture, urban planning, balanced development, water resource management, disaster management, monitoring of land encroachments, land degradation, and overall environmental health. India being a densely populated country, understanding LULC patterns is essential for efficient use of land resources. In rapidly developing areas such as Hyderabad and surrounding regions in Telangana, land use is undergoing numerous changes due to urban expansion, agricultural modernization, and infrastructural development. Traditional field surveys and obsolete maps are time consuming, resource intensive and insufficient to meet the growing demand for real-time, precise, and scalable solutions. While satellite-based classification provides a reliable approach, its effective use requires advanced tools and techniques for image preprocessing, feature extraction, classification, and spatial analysis. This research project aims to develop a Machine Learning integrated Remote Sensing framework by leveraging advanced geospatial technologies such as GEE and QGIS tools. The applications are known to provide powerful capabilities essential for satellite imagery. The research work has been able to accomplish the LULC classification into various land categories by generating zonal maps, leveraging robust machine learning algorithm i.e., Random Forest classifier with appreciable metrics such as overall accuracy, kappa coefficient, F-1 score, and zonal statistics. The project is also able to successfully experiment a novel approach by deploying KAN layers for feature extraction that aided in improved classification results. Along with these, the findings include multi-temporal spatial analysis for change detection and understanding of land dynamics.

2. KEYWORDS

Geospatial Technologies, Image Preprocessing, KAN Layers, Machine Learning, Random Forest, Remote Sensing, Zonal Maps, Zonal Statistics

3. PROBLEM STATEMENT

To optimally utilize land resources for growing population, nations require a scientific, data driven understanding of its land resources, land use and land cover classification, and the landscape dynamics. The goal is to make proper & sustainable utilization of land with holistic view of the planet's benefit, and to take balanced and informed decisions. Such a research work analysis can be further used to assess the characteristics of land, its current state and use, to generate environmental impact assessment dashboards for the stakeholders to better understand the scale and scope of implementing the development projects. The stakeholders can be climate change analysts, scientists, urban planners, construction engineers, government executives, decision makers, researchers etc. The research work is intended to achieve LULC classification with maximum efficiency, using remote sensing data for the study area of interest, and to interpret the spatial and temporal changes, considering the satellite data from a time-period of interest. A novel approach is investigated to capture non-linear data characteristics, where initial feature extraction is performed using KAN, an emerging deep learning technique expected to outperform conventional MLP layers. These enhanced features are subsequently classified using RF classifier for supervised classification, which is proved to be yielding results with commendable efficiency in multiple previous research works in this field. This framework can be further extended to support reliable land-use change prediction and planning.

The Sentinel 2 satellite optical imagery is obtained for the chosen study area i.e., Hyderabad and its surrounding regions in Telangana, ranging over years from 2019 to 2024. The satellite images are available as unstructured data, from Copernicus data hub via GEE accessibility to Sentinel satellite data. The success and accomplishment of this research project rely on precise differentiation of complex classes which is determined by the classification accuracy using reliable validations that include confusion matrices, kappa coefficient, ground truth data, generation of LULC classification maps, zonal statistics calculation and comprehending temporal dynamics between two selected years of data. While robust algorithms and advanced geospatial tools play crucial role in achieving outcome, the attainability of the accuracy also equally relies on proper preprocessing, feature extraction, texture analysis, band stacking and vegetation indices calculations, which remarkably aids in attaining desired classification accuracy.

4. ABOUT STUDY AREA

The study area of interest for this research work encompasses Hyderabad and its surrounding landscape, within an approximate radius of 100 km, with Charminar taken as reference point. The region exhibits a heterogeneous mix of land use and land cover types, including dense urban settlements, peri-urban development zones, agricultural lands, forest patches, inland water bodies, and open lands, making it highly suitable for LULC classification and zoning studies. It is known to be under a semi-arid tropical climatic zone, with a topography predominantly constituting low hillocks

and plains, while also containing numerous lakes and reservoirs that support urban water supply, agriculture and ecosystems. The city and its peri-urban surroundings have been experiencing rapid increase in population density, due to an increase in occupants via migration from districts and states, seeking better life and growth. This has led to significant land transformations such as increase in built-up areas, industrial zones etc. Proper understanding of this landscape and its surroundings dynamics is hence essential to understand the scope for balanced growth, sustainable development, distributed population and living zones.

5. LITERATURE REVIEW

Satellite-based remote sensing has become a standard technique for land classification due to its wide coverage and high temporal resolution. The approach of this research work [1] re-iterates the notion that preprocessing enhancements, coupled with robust machine learning algorithms, can significantly improve the accuracy of land use classification. The findings of another study [2] show the importance of leveraging machine learning techniques, particularly the RF classifier, in the field of remote sensing. Advanced ML techniques such as MLE, RF, and SVM, when integrated with geospatial tools, have proven effective in distinguishing built-up regions, vegetation, water bodies, and barren lands. There are extensive and valuable research works that demonstrated the productive use of modern GIS systems, RS technologies, and ML models to examine LULC dynamics. Spatial analysis was conducted in this research work [3] on Sentinel-2A imagery by using SNAP and ArcGIS software platforms, wherein, the vector files containing the training samples, exported from GEE was used for supervised classification. The SNAP toolbox is also widely used for atmospheric correction (Sen2Cor), speckle filtering, radiometric calibration, image stacking, and supervised classification. GIS and RS techniques were used in another study [4] to quantify and understand LULC changes in their study area across a 15-year period, making this study's technique simple and inexpensive.

Significant studies suggest application of GEE capabilities for machine learning based LULC analysis.[5] The productive research work executed in the paper [6] presented a systematic comparison of all supervised machine learning classifiers within the GEE platform, such as SVM, Minimum Distance Classifier (MDC), Classification and Regression Trees (CART), Gradient Tree Boosting (GTB), K-Nearest Neighbours (KNN), Naive Bayes (NB), and RF algorithms, whose results indicate the algorithms performance and variations for different land categories. Another study [7] highlighted the importance of GEE framework and the effectiveness of machine learning classifiers, especially RF and SVM, in accurately mapping LULC patterns in their study area, suggesting RF's potential as a reliable tool for remote sensing-based LULC mapping. A research case study of Telangana state agglomeration [8] proposed a spatio-temporal analysis of LULC changes, where Landsat satellite images were processed using RS, GIS, and the Object-based Image Analysis (OBIA) approach to produce LULC maps for 1990, 2000, 2010, and 2023, investigating, multiple classes as forest, agricultural land, built-up land, shrubland, wasteland and waterbodies. Similarly, a research work [9] explored LULC changes within Telangana, using QGIS MOLUSCE plugin which demonstrated the application of MLP to determine LULC changes, with significant overall Kappa coefficient value (0.78) and accuracy (82%). While several studies have mapped parts of Telangana, research focused on automated LULC classification for regional level analysis is of great importance and highly required for localized mapping, zoning and change detection using modern geospatial technologies.

6. RESEARCH OBJECTIVES

The primary objectives of this research work include:

- The collection & preprocessing of Sentinel-2 satellite imagery.
- To explore a novel approach to capture the non-linear characteristics by deploying KAN neural layers & demonstrate a hybrid ML approach for LULC analysis.
- To perform simplified LULC classification using machine learning algorithms such as supervised Random Forest classifier.
- To perform image-analysis & evaluate classification accuracy using confusion matrices, validation samples and accuracy assessment results.
- To prepare annual zoning maps (built-up, vegetation, water bodies) derived from the classification results.

Additionally, the secondary objectives of the project work constitute:

- Analysis of spatial land use changes by deploying multi-temporal satellite imagery.
- To generate geospatial outputs suitable for integration into GIS platforms for planning and policymaking.

7. METHODOLOGY

This study follows a machine learning based geospatial workflow for the LULC classification and zoning using multi-temporal Sentinel satellite imagery. The methodology is formulated to achieve optimized performance on Sentinel datasets while ensuring classification accuracy & computational efficiency. The overall framework integrates satellite data preprocessing, feature extraction, supervised machine learning classification, GIS-based zoning & further analysis.

7.1 Study Area Definition

The study area covers Hyderabad region & the surrounding landscape, defined using an AOI with a radius of approximately 100 km, taking Charminar (Latitude: 17.3616° N, Longitude: 78.4747° E) as the geographic central reference point. To precisely delineate the region of interest, administrative boundary shapefiles are used. This region is chosen due to rapid urban growth, availability of high-quality sentinel-2 imagery, presence of distributed vegetation and inland water bodies such as Hussain Sagar and others.

7.2 Tools and Software Used

- Google Earth Engine for data access and preprocessing
- QGIS for spatial analysis and visualization
- ESA World Cover data for ground truth
- Python for machine learning implementation and data analytics (Google Colab, Jupyter lab)
- A computing system with descent RAM and processing capabilities

7.3 Satellite Data Acquisition

Sentinel data refers to Earth Observation data from European Space Agency's Copernicus Sentinel satellites. Sentinel-2 Level-2A multispectral surface reflectance data was utilized for this study, covering the period from 2019 to 2024. Sentinel-2 optical imagery provides high spatial resolution (10 m) and temporal resolution (5 days), and includes spectral bands suitable for urban, vegetation & water classification. They are known to be suitable for land cover analysis as they provide atmospherically corrected surface reflectance data. Data with minimal cloud cover were chosen to ensure data quality across temporal datasets. To maintain seasonal consistency, imagery from similar months was selected wherever possible to reduce seasonal and illumination differences. To ensure inter-annual homogeneity, only core Sentinel-2 reflectance bands were retained prior to temporal compositing, avoiding any inconsistencies.

7.4 Pre-processing of Satellite Data

Pre-processing was carried out to ensure radiometric, geometric, and spatial consistency across all datasets. As part of the image preprocessing, certain steps were performed using GEE and QGIS, such as cloud shadow masking, verification of Level-2A atmospheric correction to minimize atmospheric distortions, band selection, stacking of relevant multispectral bands and resampling to a common spatial resolution. Extraction of the Region of Interest using the AOI shapefile was done, to limit analysis to the study area, by clipping to the defined AOI. Along with these, reprojection was done to a common CRS (EPSG:32644 – WGS 84 / UTM Zone 44N).

The study area boundary was imported into GEE as an asset for feature collection, and it was referenced directly using its asset ID. All the preprocessing, feature engineering, and raster files export were implemented in GEE. Each year's imagery was prepared as a multi-band raster containing identical spectral information to ensure compatibility during classification & change analysis. Due to export tiling constraints in GEE, yearly Sentinel-2 composites were initially generated as multiple tiles. These tiles were mosaicked in QGIS using GDAL merge operation to produce a single seamless raster per year prior to training data extraction and classification. Below table provides an idea about Sentinel-2 bands and their resolutions. ^[1]

Table 1 Sentinel-2 band – Resolution

Band 1 (B1)	Coastal aerosol 60 m
Band 2 (B2)	Blue 10 m
Band 3 (B3)	Green 10 m
Band 4 (B4)	Red 10 m
Band 5 (B5)	Vegetation Red Edge 20 m
Band 6 (B6)	Vegetation Red Edge 20 m
Band 7 (B7)	Vegetation Red Edge 20 m
Band 8 (B8)	NIR 10 m
Band 8A (B8A)	Vegetation Red Edge 20 m
Band 9 (B9)	Water vapour 60 m
Band 10 (B10)	SWIR - Cirrus 60 m
Band 11(B11)	SWIR 20 m
Band 12 (B12)	SWIR 20 m



Figure 1 Sentinel-2 GeoTIFF raster file for the AOI (2019)



Figure 2 Sentinel-2 GeoTIFF raster file for the AOI (2021)

7.5 Feature Engineering and Spectral Indices

Sentinel-2 Level-2A surface reflectance imagery from October to December 2022 was used to derive predictor variables. This post-monsoon period was selected to minimize cloud contamination and enhance spectral separability among land cover classes. Cloud and shadow masking was performed using the Scene Classification Layer (SCL), and a median composite was generated to further suppress residual noise.

Spectral bands representing visible and near infrared regions (B2, B3, B4, B8, B11, B12) were selected to ensure homogeneity across the dataset. Additionally, three commonly used spectral indices were computed and appended to the feature stack to enhance class discrimination, improve class separability and overall classification accuracy.

- NDVI for vegetation detection
- NDWI for water body identification
- NDBI for urban area detection

These indices were stacked along with spectral bands to form a comprehensive and consistent feature space for classification of each year.

7.6 Ground Truth Data and Training Sample Preparation

Manual digitization of training samples was avoided to reduce subjectivity and improve reproducibility. Instead, training labels were derived from the ESA World Cover entity, a globally validated and automatic reference land-cover dataset, enabling automated, unbiased sampling for supervised classification. The ESA World Cover 2021 global land cover product was used as the reference dataset for generating training labels. It provides a spatial resolution of 10 m and is derived using Sentinel-1 and Sentinel-2 data, with an overall reported accuracy of approximately 80%. Owing to its global consistency and independent validation, it is widely adopted as a pseudo ground truth source in large-area LULC studies where field-collected ground truth data are hard to be available.

Although the reference land cover data correspond to the year 2021 & the Sentinel-2 predictor variables were derived from 2022 imagery, this temporal mismatch is considered acceptable for the present study. Major land cover classes such as urban areas, permanent water bodies & large agricultural regions typically exhibit strong temporal persistence over short time intervals. Under the assumption of temporal stability of dominant land cover types, World Cover 2021 serves as a reliable source of training labels for mapping LULC in 2022. This cross-year training strategy is commonly adopted in remote sensing & ML based LULC studies, particularly when recent, validated ground truth data are sparsely available. The approach supports generation of up-to-date land cover maps while maintaining label reliability.

Only cropland, built-up & permanent water classes were retained and reclassified to match the study objectives. Specifically, cropland (class code 40) was mapped to agriculture, built-up areas (class code 50) to urban, and permanent water bodies (class code 80) to water. All other classes were masked out to ensure that only spectrally & semantically reliable categories were included in the training process. This reclassification reduces class ambiguity, minimizes spectral overlap & improves classification robustness. Thus, World Cover class labels were re-mapped into three project-specific classes.

Table 2 Land Use Class Labels

Class ID	Class Label	ESA World Cover Definition
1	Built-up	Built-up Areas (50)
2	Vegetation	Cropland (40)
3	Water bodies	Permanent Water (80)

These labels were used to generate pixel-level training samples for supervised machine learning. Stratified random sampling was employed to generate balanced training datasets, ensuring equal representation of agricultural, urban, and water classes. The balanced training samples constituted 1500 samples per class, resulting in a total of 4500 training points, at a spatial resolution of 10 m, which is a good number for training samples suitable for the methodology, and

for a 3-class LULC study. Class balance was maintained to mitigate class bias in the supervised learning process and to improve the generalization ability of supervised learning models. The year 2022 was selected due to minimal cloud cover and representative land surface conditions. The trained model was applied consistently across all study years.

7.7 Model Training and Classification Using Machine Learning

A three-class LULC design (Built-up, Water Bodies, Vegetation) was adopted to ensure spectral separability, temporal consistency, and reliable inter-annual comparison across the study period. Stratified random samples of multi-year sentinel-2 data were used to derive temporal statistical features, to capture land cover stability and dynamics. The extracted feature dataset was used as input for a hybrid ML framework. The learning pipeline was implemented, where KAN was first employed to learn non-linear feature representations from Sentinel-2 spectral and index features. It models the complex nonlinear relationships through adaptive functional transformations, thus enhancing feature representations. These transformed features were subsequently classified using RF classifier for supervised classification and to generate nominal class labels. Thus, a hybrid and novel ML framework combining KAN for nonlinear feature extraction and RF for supervised classification was employed. The RF classifier was selected due to its robustness to noise, ability to handle high-dimensional data, resistance to overfitting, and proven effectiveness in LULC classification tasks. This combination leverages the feature learning capability of KAN and the ensemble decision-making strength of RF, resulting in improved classification accuracy and interpretability.

ML model training was conducted in Google Colab, leveraging cloud-based Python environment for reproducibility and computational efficiency. Model artifacts and datasets were stored in Google Drive to ensure consistency. The model was trained using the 2022 feature stack, as this year represented stable and diverse land cover conditions. The trained model was downloaded & then applied independently to classify each year (raster file) from 2019 to 2024. Classification code was executed in python using Jupyter lab. Each classified output was generated as a categorical raster file (.tiff file) representing the three LULC classes. Considering the possible computational constraints, annual land cover classification was performed sequentially for each year using a fixed trained model, ensuring consistency in model parameters and class definitions across all years.

Figure 3 shows the flowchart of the methodology and workflow of the project, by sub-dividing major tasks into different logical phases of the study.

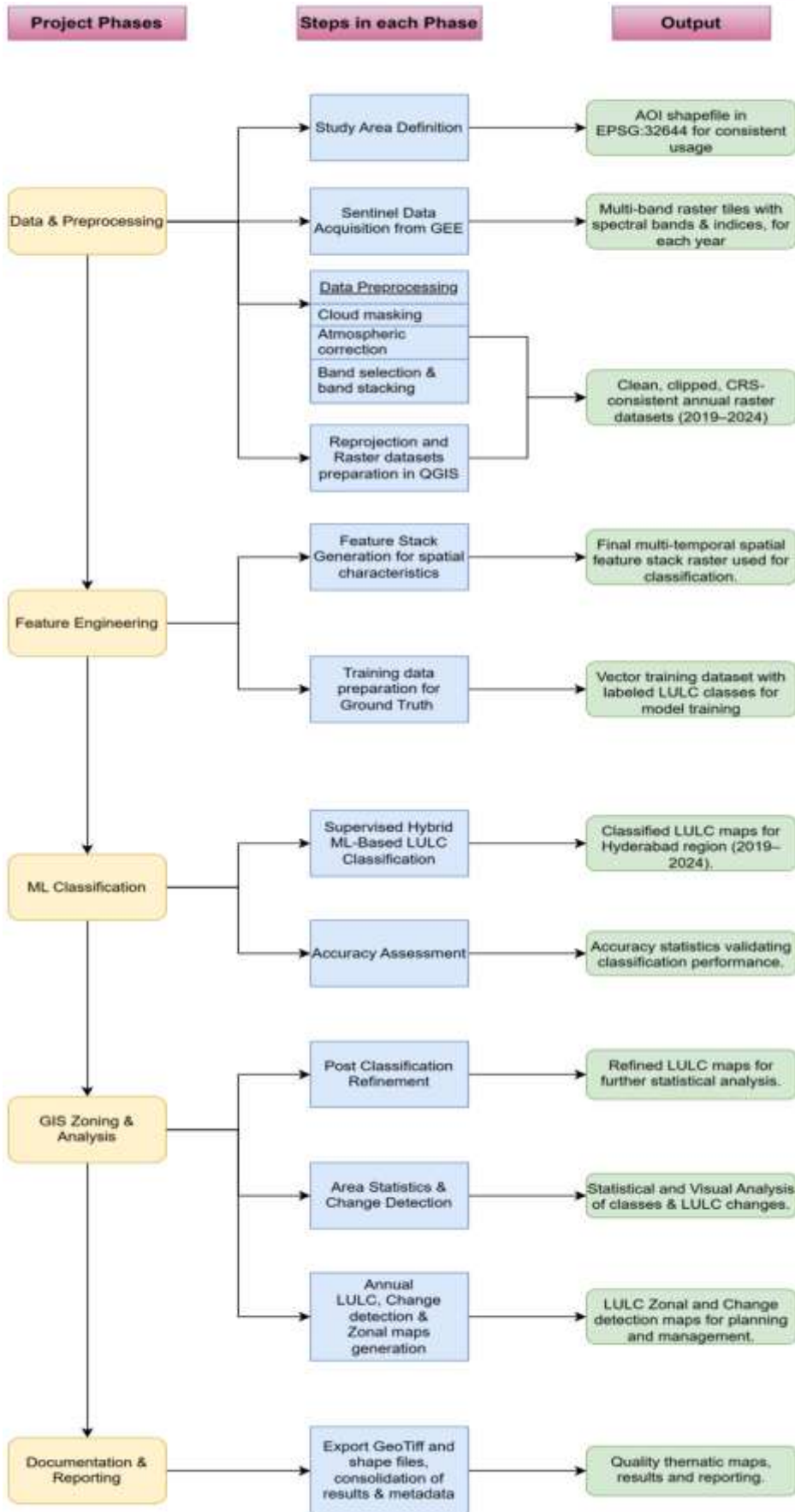


Figure 3 Methodology workflow of the study

7.8 GIS-Based Zoning and Change Detection

The classified LULC outputs can be refined further and integrated into GIS platforms (QGIS) for zoning analysis. Zoning maps were generated to identify and delineate major functional zones in scope, such as vegetation, water and built-up areas. Multi-temporal classified datasets were analysed to perform change detection, enabling assessment of land use transitions and urban expansion trends over the 2019-2024 period.

7.9 Post-classification Refinement

The classified LULC raster is a single-band thematic dataset where each pixel stores a discrete land-use class label. Initially, the raster appears in grayscale due to default rendering in QGIS. A pseudocolor symbology was applied to visualize individual land-cover classes distinctly. Discrete integer class values were assigned standardized colors representing built-up, vegetation & water bodies. A consistent legend was applied across all temporal datasets to ensure accurate visual comparison.

Initial classification results showed water misclassification in certain years. Certain processing steps were performed such as NDWI-based water masks were generated, threshold-based refinement was applied, and raster calculator logic was used to enforce water class dominance where NDWI indicated true water. This step significantly reduced false positives and improved temporal consistency of water bodies such as Hussain Sagar Lake.

Since ML classifiers assign arbitrary numeric labels to output classes, visual interpretation was performed using known reference features and dense urban regions to correctly map raster values to land-use classes prior to legend assignment. This process ensured accurate assignment of land-use class names to raster values before the legend creation.

7.10 Raster Alignment

To enable accurate pixel-wise comparison across years, all classified rasters were aligned to a single reference grid (LULC 2019). Raster alignment ensured identical CRS, same pixel size, same spatial extent, and same grid origin. Nearest-neighbour resampling was used to preserve categorical class values.

7.11 Area Estimation and Statistics

To support planning & zoning applications, zonal statistics are computed to summarize land cover distributions within administrative units. This enables spatial comparison & quantitative assessment of land use patterns across study region. Post-classification analysis was performed to compute preliminary land-use area statistics for each class within the study area. Temporal changes were quantified to assess urban expansion and land-cover dynamics. Although the input satellite imagery was initially downloaded for the study area, a final AOI-based raster masking was performed on the classified rasters, to ensure that only valid pixels within the study boundary were considered during area estimation. Pixel counts were converted to area using spatial resolution and pixel-based area calculations were conducted using unique value counts. No-Data pixels were excluded from all calculations to avoid distortion of results. The final class-wise area statistics were computed year-wise, for all LULC classes using raster histograms.

7.12 LULC Change Detection

Change detection was performed using raster algebra and cross-tabulation of aligned LULC maps. Change matrices were generated to quantify transitions between classes, enabling analysis of urban expansion, vegetation loss or gain, water body changes. This provided a comprehensive temporal understanding of land cover dynamics in the study area.

8. RESULTS AND ANALYSIS

8.1 Accuracy Assessment

Accuracy assessment is a critical component of any LULC classification studies, as it quantitatively evaluates the reliability and predictive performance of the classification model. In this study, the accuracy of the classification framework was evaluated using standard accuracy assessment metrics, based on validation samples independent of

training data. A confusion matrix was generated to compute precision, recall, overall accuracy & the Kappa coefficient. Additionally, F1-scores were calculated to evaluate the balance between precision and recall for each LULC class.

KAN model training execution phases:

```
kan.fit(
    dataset,
    steps=200
)
```

... | train_loss: 2.14e-02 | test_loss: 4.76e-01 | reg: 1.66e+02 | : 100% | 200/200 [16:56<00:00, 5.08s] saving model version 0.1

```
{'train_loss': [array(0.17428052, dtype=float32),
array(0.15223755, dtype=float32),
array(0.1391841, dtype=float32),
array(0.13023527, dtype=float32),
array(0.12190706, dtype=float32),
array(0.13965763, dtype=float32),
array(0.128885, dtype=float32),
array(0.11893366, dtype=float32),
array(0.1176993, dtype=float32),
array(0.1176993, dtype=float32)],
'test_loss': [array(0.16173601, dtype=float32),
array(0.15391469, dtype=float32),
array(0.15725416, dtype=float32),
array(0.16179518, dtype=float32),
array(0.16469945, dtype=float32),
array(0.1545511, dtype=float32),
array(0.15831524, dtype=float32),
array(0.15831524, dtype=float32),
array(0.15831524, dtype=float32),
array(0.15831524, dtype=float32)]}
```

Random Forest Classifier execution:

```
RandomForestClassifier(
    max_depth=20, n_estimators=300, n_jobs=-1,
    random_state=42)
```

Accuracy Metrics execution output:

```
[[409 38 3]
 [ 28 420 2]
 [ 6 3 441]]
```

	precision	recall	f1-score	support
1	0.92	0.91	0.92	450
2	0.91	0.93	0.92	450
3	0.99	0.98	0.98	450
accuracy			0.94	1350
macro avg	0.94	0.94	0.94	1350
weighted avg	0.94	0.94	0.94	1350

8.1.1 Confusion Matrix Analysis

The confusion matrix obtained from the validation dataset is presented as shown below:

Reference / Predicted	Class 1	Class 2	Class 3
Class 1	409	38	3
Class 2	28	420	2
Class 3	6	3	441

Here, Class 1 represents Built-up land, Class 2 represents Vegetation, Class 3 represents Water Bodies

The diagonal elements indicate correctly classified samples, while off-diagonal elements represent misclassifications. The confusion matrix reveals a high number of correctly classified samples across all classes, with minimal confusion between water bodies and other land cover types. Minor misclassification is observed primarily between built-up and vegetation classes, which is expected due to mixed spectral characteristics in transitional urban and vegetation zones.

8.1.2 Overall Accuracy

Overall Accuracy of the classification is calculated as the ratio of correctly classified samples to the total number of validation samples. An overall accuracy of 94% is obtained upon evaluation of the model, which indicates that the model performs exceptionally well in distinguishing between the three major land cover classes across the study area.

8.1.3 Class-wise Precision, Recall, and F1-score

To further evaluate class-level performance, precision, recall, and F1-score were computed for each land cover class, as summarized below.

Class	Precision	Recall	F1-score
Built-up (Class 1)	0.92	0.91	0.92
Vegetation (Class 2)	0.91	0.93	0.92
Water (Class 3)	0.99	0.98	0.98

Precision measures the correctness of predicted samples for a given class, while recall indicates how well the model captures all reference samples of that class. The F1-score is the harmonic mean of precision & recall, which provides a balanced measure of classification performance. It can be noticed that the water class exhibit the highest precision (0.99) and recall (0.98). Built-up and vegetation classes also show consistently high F1-scores (0.92), indicating reliable discrimination despite spectral overlap commonly encountered in urban and peri-urban environments. This confirms the effectiveness of spectral features and derived indices used during training, as well as the robustness of KAN-based feature extraction and RF classification. It demonstrates the model's strong ability to accurately delineate the classes.

8.1.4. Kappa Coefficient

Cohen's Kappa coefficient was computed to assess the agreement between classified output & reference data. Kappa > 0.80 is generally interpreted as strong agreement. The obtained Kappa coefficient of ~0.91 confirms that the classification accuracy reflects a high level of consistency between predicted and reference land cover classes.

```
# Compute kappa
kappa = cohen_kappa_score(y_true, y_pred)
print("Cohen's Kappa:", round(kappa, 3))

*** Cohen's Kappa: 0.911
```

Table 3 Consolidated Accuracy Assessment Results

Model	Year	Class	Overall Accuracy %	Precision	Recall	F-1	Kappa Value
KAN (feature extractor) + RF Classifier	2022	Class 1	94	0.92	0.91	0.92	0.91
		Class 2		0.91	0.93	0.92	
		Class 3		0.99	0.98	0.98	

8.2 Classified LULC Maps

The trained model was applied to Sentinel-2 imagery for six temporal snapshots. Below are the execution snapshots, where trained models were applied on each year (2019 to 2024) & classified LULC .tiff files are generated as outputs.

```
print(f'\n► Processing year (year)')
classify_raster(input_tif, output_tif)
```

Processed 189,000,000 / 197,489,399 pixels	Processed 192,500,000 / 197,489,902 pixels	2026-01-19 20:08:13.270656
Processed 189,500,000 / 197,489,399 pixels	Processed 193,000,000 / 197,489,902 pixels	2026-01-19 20:08:27.067284
Processed 190,000,000 / 197,489,399 pixels	Processed 193,500,000 / 197,489,902 pixels	2026-01-19 20:08:40.712495
Processed 190,500,000 / 197,489,399 pixels	Processed 194,000,000 / 197,489,902 pixels	2026-01-19 20:08:54.537041
Processed 191,000,000 / 197,489,399 pixels	Processed 194,500,000 / 197,489,902 pixels	2026-01-19 20:09:08.120764
Processed 191,500,000 / 197,489,399 pixels	Processed 195,000,000 / 197,489,902 pixels	2026-01-19 20:09:21.982185
Processed 192,000,000 / 197,489,399 pixels	Processed 195,500,000 / 197,489,902 pixels	2026-01-19 20:09:35.750684
Processed 192,500,000 / 197,489,399 pixels	Processed 196,000,000 / 197,489,902 pixels	2026-01-19 20:09:49.566978
Processed 193,000,000 / 197,489,399 pixels	Processed 196,500,000 / 197,489,902 pixels	2026-01-19 20:10:03.418006
Processed 193,500,000 / 197,489,399 pixels	Processed 197,000,000 / 197,489,902 pixels	2026-01-19 20:10:17.010265
Processed 194,000,000 / 197,489,399 pixels	✓ Saved: LULC_2024.tif	
Processed 194,500,000 / 197,489,399 pixels	2026-01-19 20:10:31.548784	
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✓ Saved: LULC_2019.tif		

For detailed visual interpretation, classified maps corresponding to 2019 & 2021 were presented in study. Outputs depict three dominant land cover classes: Built-up, Vegetation & Water Bodies. The spatial distribution patterns observed in both years highlight the study area's rapid built-up expansion around urban core, while vegetation & agricultural land exhibits gradual fragmentation, and major water bodies such as Hussain Sagar remain spatially consistent after post-classification refinement with noticeable changes in surface water extent.

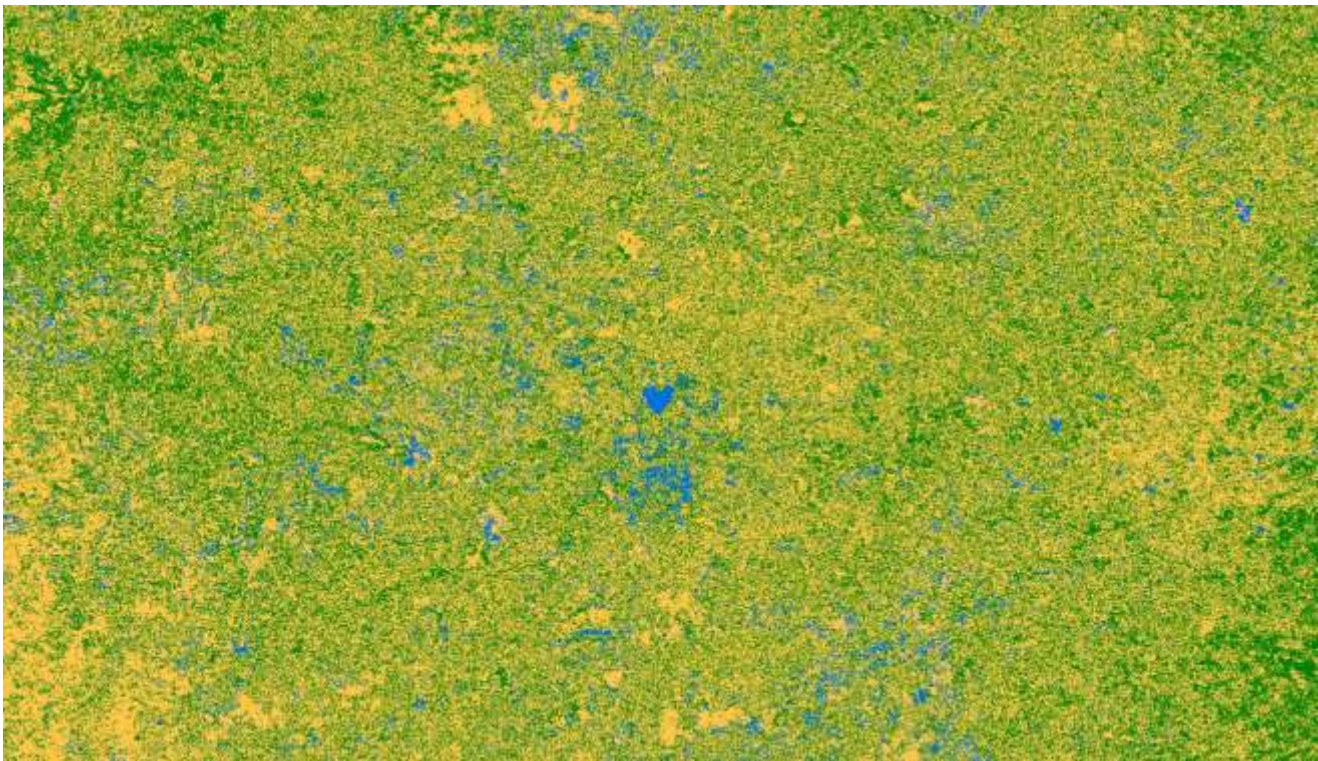


Figure 4 LULC Map of Hyderabad Region (2019)

Land Use Land Cover Map of Hyderabad (2019)

Projection: WGS 84 / UTM Zone 44N
(EPSG:32644)
Data Source: Sentinel-2, ESA WorldCover
Classification: KAN + Random Forest

Band 1 (Gray)
Built-up
Water Bodies
Vegetation

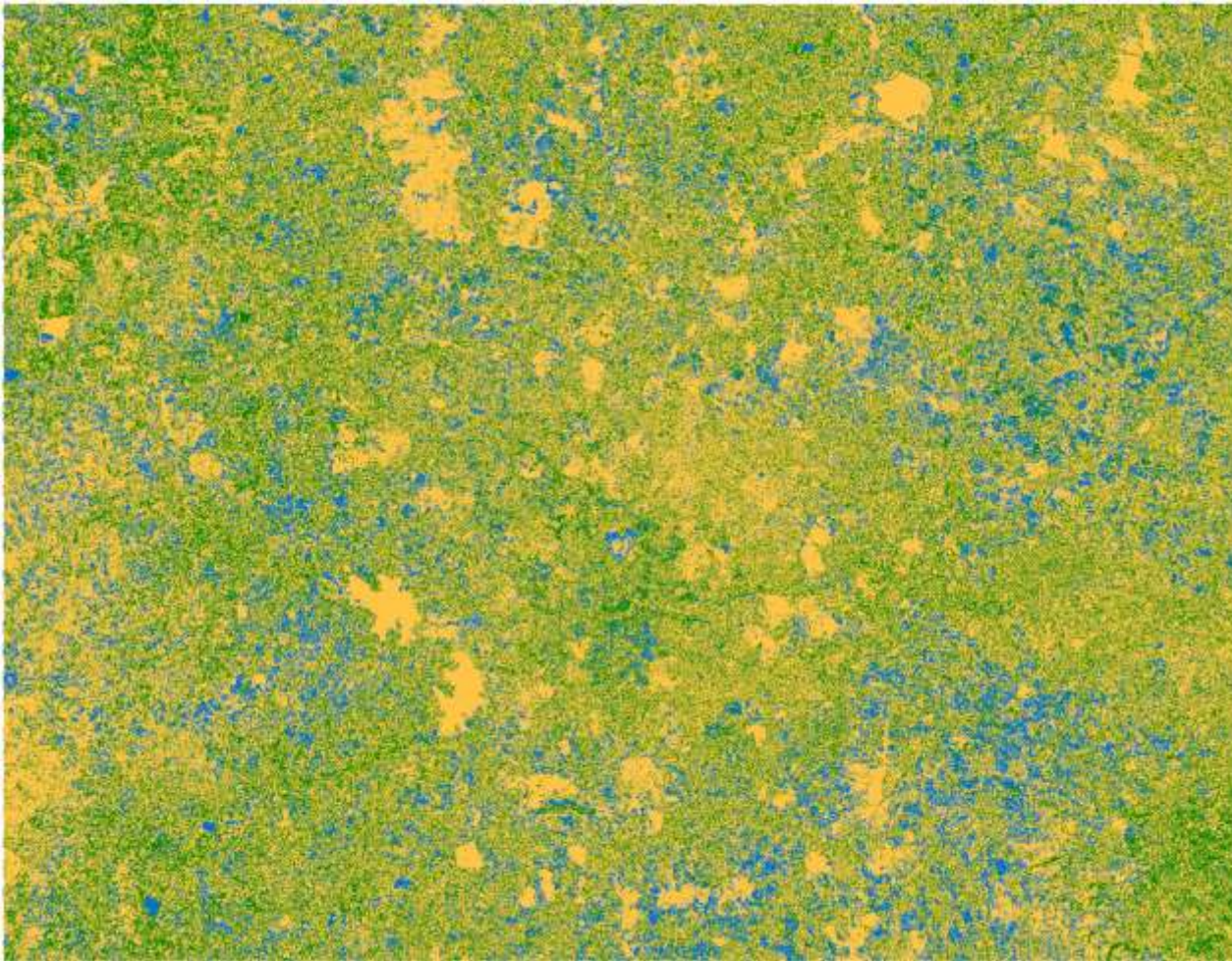


Figure 5 LULC Map of Hyderabad Region (2021)

Land Use Land Cover Map of Hyderabad (2021)

Projection: WGS 84 / UTM Zone 44N
(EPSG:32644)
Data Source: Sentinel-2, ESA WorldCover
Classification: KAN + Random Forest

Band 1 (Gray)
Built-up
Water Bodies
Vegetation

8.3 Temporal Consistency Analysis

All classified rasters were aligned to a common reference grid to ensure pixel-wise comparability. This alignment eliminated spatial shifts which may be caused by minor differences in acquisition geometry, resampling during preprocessing, variation in raster extents. After alignment, temporal comparisons showed consistent spatial correspondence, allowing reliable change detection analysis.

8.4 Area Statistics Analysis

Class-wise area statistics were computed for each year using raster histogram analysis. No-Data pixels were excluded to avoid distortion of results. Quantitative assessment of land cover distribution was performed using regional area statistics derived from the classified rasters. The results are summarized in below table.

Table 4 LULC Area Statistics for Hyderabad AOI

Year	Built-up (km ²)	Water (km ²)	Vegetation (km ²)	Total (km ²)	Built-up (%)	Water (%)	Vegetation (%)
2019	385.489	68.758	257.878	712.125	54.132	9.655	36.212
2021	406.678	85.442	220.001	712.121	57.108	11.998	30.894

Between 2019 & 2021, built-up land increased by approximately 21.19 km², reflecting rapid urban development. Vegetation experienced a substantial decline of 37.88 km², indicating conversion of green spaces into urban land uses. Conversely, water bodies increased by 16.68 km², which may be attributed to seasonal variability, reservoir expansion, or improved water detection through spectral indices.

The proportional increase in built-up area from 54.13% to 57.11% highlights the dominance of human-centred land use in the region. The reduction in vegetation cover raises concerns regarding environmental sustainability, and ecological balance. These quantitative trends corroborate the visual interpretation of the classified maps.

8.5 Change Detection Analysis

A pixel-based post-classification comparison was performed to assess land cover transitions between 2019 and 2021. The change detection matrix is presented in Table 5.

Table 5 LULC Change Detection Matrix (2019 → 2021)

From	To	Code	Area (sq.km)
Built-up	Built-up	11	52.341
Built-up	Water	12	5.5957
Built-up	Vegetation	13	24.338
Water	Built-up	21	7.1656
Water	Water	22	9.0294
Water	Vegetation	23	6.0281
Vegetation	Built-up	31	26.95
Vegetation	Water	32	5.0761
Vegetation	Vegetation	33	26.132

Change detection results reveal that the most dominant transition is from Vegetation to Built-up (26.95 km²), confirming significant urban encroachment. The persistence of built-up land (52.34 km²) indicates stable urban zones, while transitions between vegetation and water highlight dynamic hydrological conditions and land management practices.

A dedicated change detection map (2019-2021) visually illustrates these transitions, making spatial patterns of urban expansion & environmental change evident.

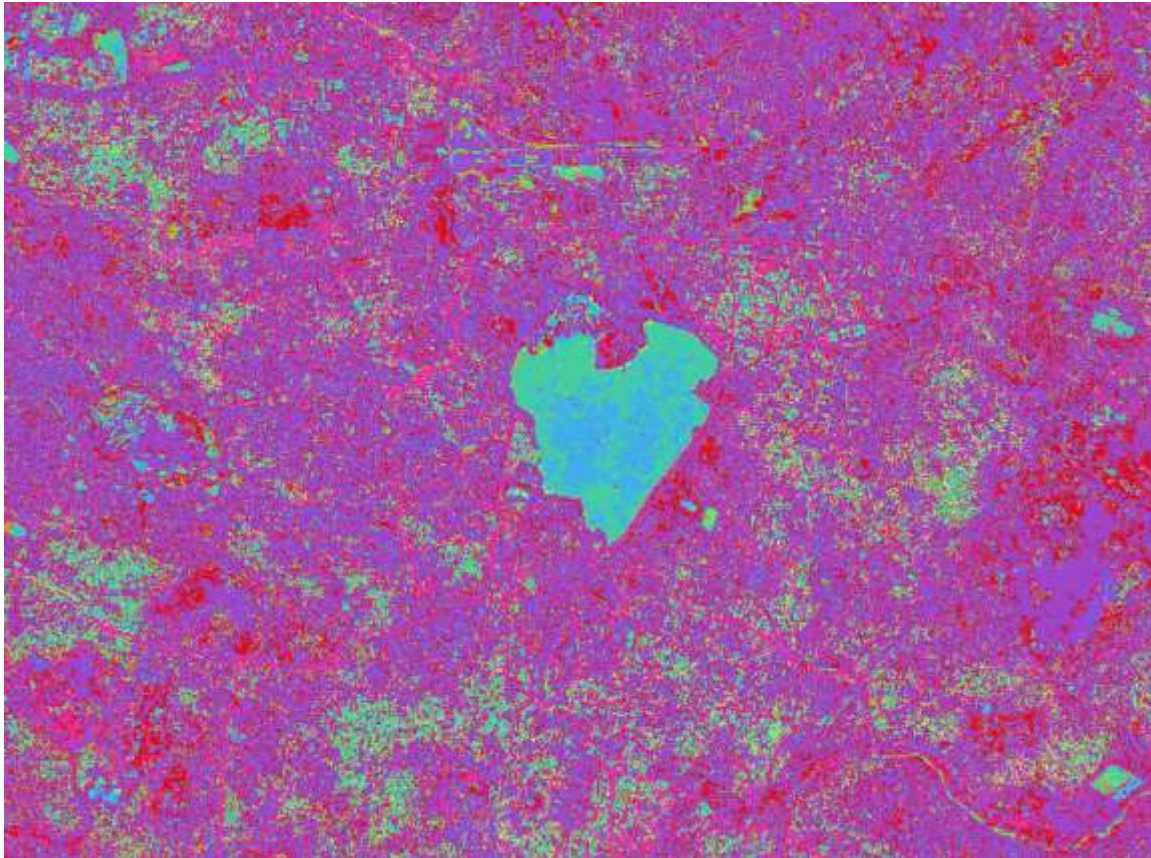
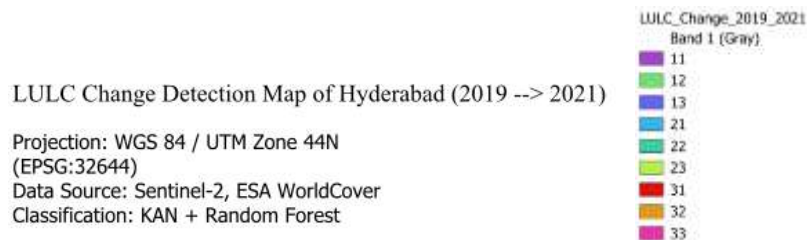


Figure 6 LULC Change Detection Map of Hyderabad (2019-2021)



8.6 Visual Analysis of LULC Changes

The bar chart in figure 7 illustrates the distribution of major land use-land cover classes constituting Built-up, Water, and Vegetation, for Hyderabad region during 2019 & 2021. The results indicate that built-up land constitutes the dominant class in both years, occupying approximately 385.49 km² (54.13%) in 2019 and increasing to 406.68 km² (57.11%) by 2021. Water bodies show a noticeable increase from 68.76 km² (9.66%) to 85.44 km² (12.00%), which may be attributed to seasonal variability, improved detection, or hydrological changes. In contrast, vegetation cover exhibits a significant decline from 257.88 km² (36.21%) in 2019 to 220.00 km² (30.89%) in 2021, highlighting ongoing land transformation. Overall, bar chart clearly demonstrates growing dominance of urban areas at the cost of vegetated land.

Figure 8 depicts the temporal trend of built-up area expansion between 2019 and 2021 using a line graph representation. The plot reveals a consistent upward trajectory in built-up land, increasing by approximately 21.19 km² over the two-year period. The observed trend reflects the increased urbanization and aligns with regional population growth, emphasizing the pressure enforced on natural land covers. The line chart effectively captures the magnitude and direction of urban expansion, reinforcing the quantitative findings derived from area statistics.

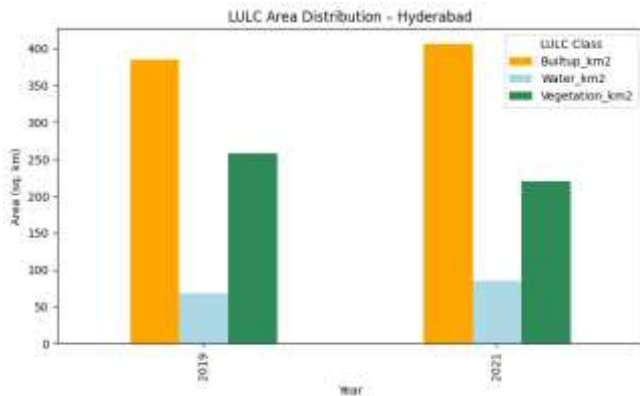


Figure 7 LULC Area Distribution (2019 & 2021)

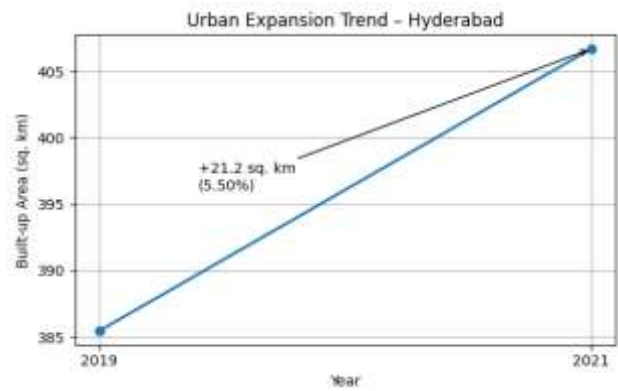


Figure 8 Urban (Built-up) Growth Trend

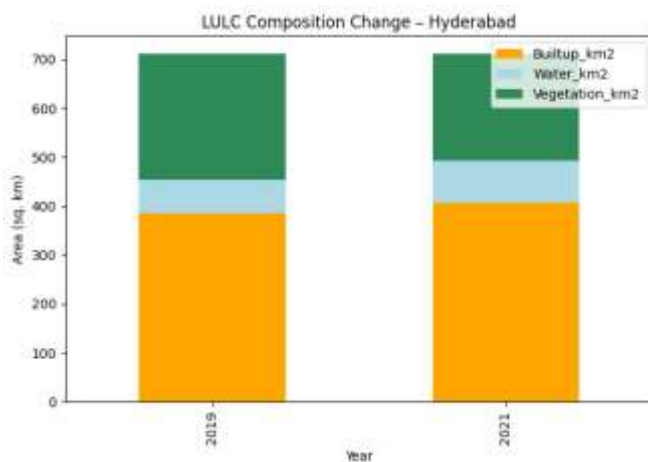


Figure 9 LULC Composition Change

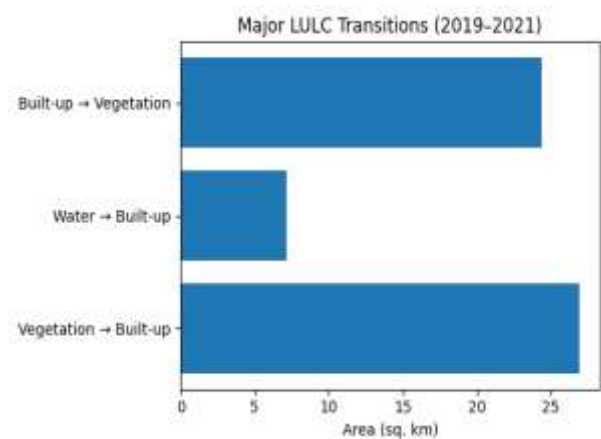


Figure 10 Change Detection Transition Plot (2019-2021)

Figure 9 illustrates the changes in the overall land cover composition between 2019 and 2021. Each bar depicts the proportional contribution of built-up land, water bodies, and vegetation to the total study area. A visible increase in the built-up component is observed in 2021, accompanied by a reduction in vegetation share, and a comparatively smaller proportion by water bodies with relative increase in 2021 is clearly identifiable in the stacked composition. This visualization provides a holistic view of how land cover composition has shifted over time, emphasizing the structural transformation of the landscape rather than absolute area changes.

Figure 10 presents a horizontal bar chart summarizing major land cover transitions between 2019 and 2021. The dominant transitions include conversion of vegetation to built-up land (26.95 km²) and water to built-up land (7.17 km²), indicating direct encroachment of urban expansion into natural land covers. Conversely, limited transitions from built-up to vegetation are also observed, suggesting any localized repurposing or possible classification uncertainty. The transition analysis highlights the dynamic nature of land use change and provides clear indication of urban growth being the primary reason for landscape transformation in the study area.

8.7 NDWI-Based Validation of Water Bodies

To validate the accuracy of water body classification, the NDWI was independently derived from Sentinel-2 imagery. Pixels with NDWI values exceeding the selected threshold were considered potential water surfaces. The NDWI-derived water mask showed strong spatial agreement with classified water bodies. However, minor false positives were observed in shadowed urban areas and moist vegetation zones, which is a known limitation of NDWI-based methods. These were mitigated through morphological filtering and contextual interpretation. This validation step confirms that the water class in the classified maps is supported by spectral evidence.

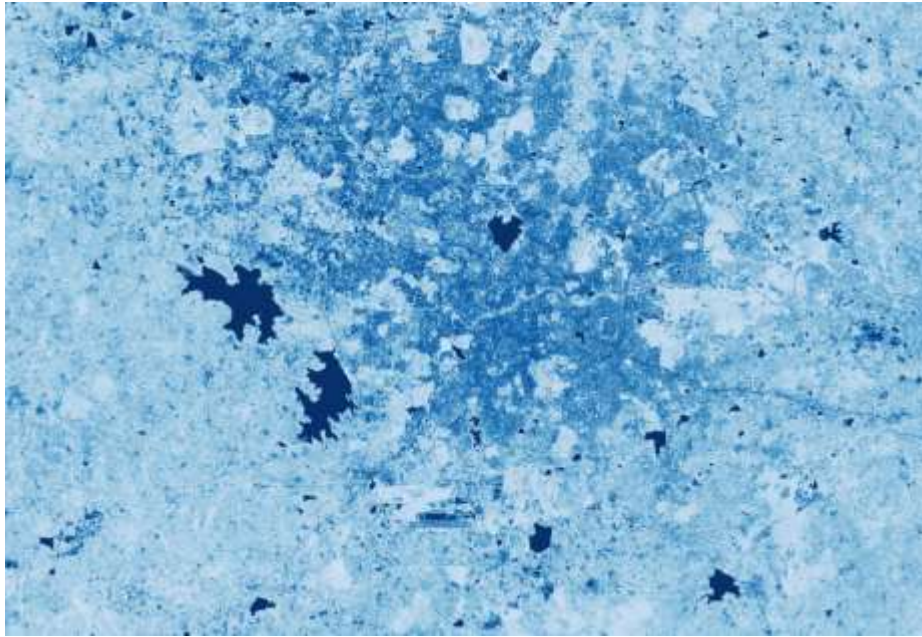


Figure 11 LULC NDWI Map of Hyderabad (2024)

LULC NDWI Map of Hyderabad (2024)

Projection: WGS 84 / UTM Zone 44N
 (EPSG:32644)
 Data Source: Sentinel-2, ESA WorldCover
 Classification: KAN + Random Forest



9. Findings

The accuracy assessment results clearly demonstrate that the hybrid KAN-RF model provides robust, reliable, and consistent classification performance for LULC mapping over the Hyderabad study region. The integration of KAN for non-linear feature extraction with Random Forest for classification enhanced the model's ability to handle complex spectral variability and mixed land cover patterns. The strong performance of Random Forest classifier is also attributed due to the balanced training samples, multi-spectral feature stack, and integration of derived spectral indices. The model achieved an overall classification accuracy of 94%. Water bodies were identified with near-perfect accuracy with an F1-score of 0.98, while agricultural and urban classes achieved F1-scores of approximately 0.92. It demonstrates the advantage of combining non-linear feature extraction with ensemble learning. It indicates that the approach is appropriate for moderate resolution satellite imagery and urban environments with complex spectral characteristics.

Stable performance for water bodies, built-up and vegetation classes ensure applicability for hydrological analysis, urban growth and environmental change assessments. Minor confusion between agriculture and urban classes is attributed to mixed pixels at the urban-rural interface, which is consistent with expectations at 10 m spatial resolution. Training the model on a single representative year and applying it to multiple years proved effective, provided that the input features were standardized, seasonal consistency was maintained and post-classification refinement was applied. These results validate the suitability of the proposed approach for multi-temporal LULC analysis using Sentinel-2 imagery.

The findings of this study highlight the significant urbanization of Hyderabad between 2019 and 2021, characterized by a notable increase in built-up areas and a corresponding decline in vegetation cover. These trends are consistent with Hyderabad's population growth, infrastructure expansion, and developmental activities. The observed increase in water bodies contrasts with vegetation loss and suggests the influence of seasonal variability, reservoir management practices, and classification sensitivity to spectral indices. The NDWI validation confirms the reliability of water classification while also emphasizing the need for careful interpretation in shadowed or mixed regions.

Nevertheless, some inconsistencies across years underline the importance of enhanced pre-processing, radiometric normalization, seasonal consistency, and adequate training sample representation. Post-classification refinement is understood to be essential for reliable multi-temporal LULC analysis. Overall, the integrated ML and GIS workflow produced consistent and interpretable results.

10. Limitations

Despite of achieving high classification accuracy and producing meaningful spatio-temporal insights, the present study has certain limitations that should be acknowledged.

Firstly, the LULC classification was constrained to three major land cover classes as Built-up, Vegetation, and Water. While this simplification ensures robustness and reduces class confusion, it does not capture finer land cover categories such as barren land, agricultural cropland, forest types, or wetlands. This may lead to generalized representations, especially in peri-urban and mixed land-use regions.

Secondly, the training samples were primarily derived from a single reference year using ESA World Cover-based labelling. This ensured consistency in class definitions, but it may not fully capture spectral variability across different years, seasons, or atmospheric conditions. As a result, certain areas, particularly water bodies and vegetated zones, exhibited temporal class inconsistencies due to seasonal changes or mixed pixel effects. Although the overall accuracy and class-wise metrics indicate strong model performance, the ground truth data being derived from automated sources rather than extensive field surveys may introduce minor labelling uncertainties, particularly in transitional land cover zones.

These limitations were mitigated where possible through refinement and alignment techniques. Future work may address these by incorporating multi-year reference datasets, field-based validation, higher temporal resolution imagery, increased number of land cover classes, and advanced change-detection techniques to further enhance robustness.

11. Future Scope

The current study lays a strong foundation for an advanced urban land cover analysis, and several enhancements can be explored in future work.

One key extension would be incorporating additional land cover classes, such as barren land, forest, agricultural land, and wetlands. Expanding the class taxonomy would enable more detailed environmental assessments and support applications such as urban sprawl analysis and land degradation studies. It can improve thematic details and subtle land cover transitions.

Future research can also benefit from multi-year and multi-season training data, allowing the model to better generalize across temporal variability. Training the model using combined datasets from multiple years would reduce seasonal bias and improve temporal consistency in classified outputs.

The adoption of object-based image analysis provides another potential direction, as segmentation of meaningful objects before classification could significantly reduce salt-and-pepper noise.

The methodology can explore other significant ML models such as SVM and MLE for comparative analysis and appropriateness for the case study. It can also be extended to long-term change detection, urban growth modelling, sub-regional analysis and scenario-oriented experimentations to support sustainable development initiatives.

12. Conclusion

This study successfully implemented a comprehensive framework for multi-temporal LULC mapping & change detection using a hybrid KAN and Random Forest approach applied to Sentinel-2 satellite imagery. Application of KAN to remote sensing and geo-spatial analysis is the novel idea, whose suitability is demonstrated in this research work.

The developed model achieved high classification accuracy, with an overall accuracy of 94% and strong class-wise performance across built-up, vegetation, and water categories. The integration of non-linear feature extraction via KAN and ensemble learning by Random Forest proved effective in capturing complex spectral patterns in heterogeneous urban landscapes.

The results revealed significant urban expansion in the Hyderabad region between 2019 and 2021, characterized by an increase in built-up areas and a corresponding decline in vegetation cover. The application of post-classification change detection supported meaningful interpretation of temporal land cover transitions.

Despite certain limitations related to class granularity, seasonal variability, and pixel-based analysis, the study provides a scalable, reproducible, and data-driven methodology suitable for urban LULC monitoring. The outcomes demonstrate the potential of combining machine learning and remote sensing techniques for informed decision-making in urban planning and environmental management.

Overall, this research work contributes a robust data-science oriented approach to geospatial analysis and serves as a valuable reference for future research in satellite image classification and land cover change studies.

13. References

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I hereby declare that all activities related to the Capstone Project including the completion of the Coursera courses and the execution of the project work have been carried out by me to the best of my knowledge.

I declare that generative AI tools were not used for this project work. I take full responsibility for the accuracy and integrity of this report.

Aishwarya Chanda